Impacts of Gun Characteristics on the Lethality of Gun Violence in the United States

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Background

- **Prevalence of Gun Violence in the US:** Gun violence is a critical issue affecting communities across the United States, necessitating research into its causes and consequences to inform policy and preventative measures.
- **Data Source and Organization:** Data is comprised of gun violence cases in the US. The 2017 dataset was specifically chosen due to its completeness.
- **Project Objective and Methodology:** The overarching goal is to explore statistical predictors of lethality in gun violence, using visualizations like bar graphs to elucidate findings regarding gun types and statuses.

Questions of Interest

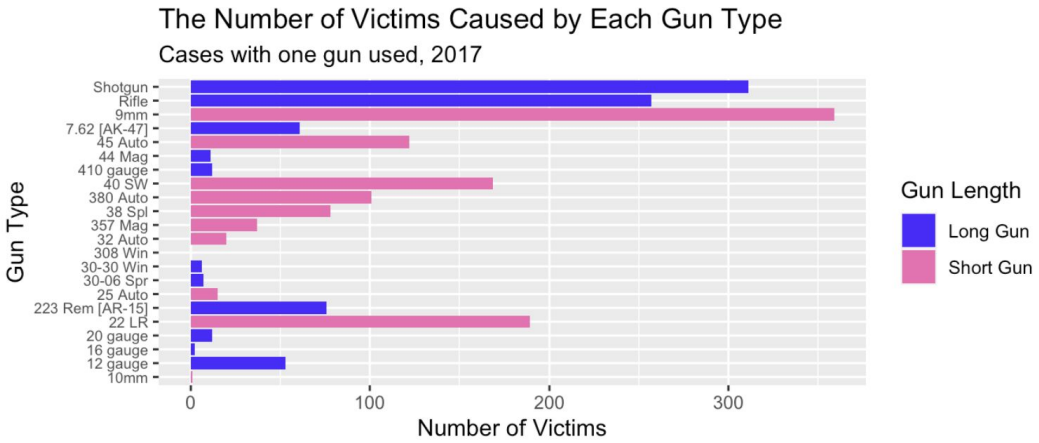
Question 1: Is there a statistically significant difference in the proportion of casualties between long barrel and short barrel guns?

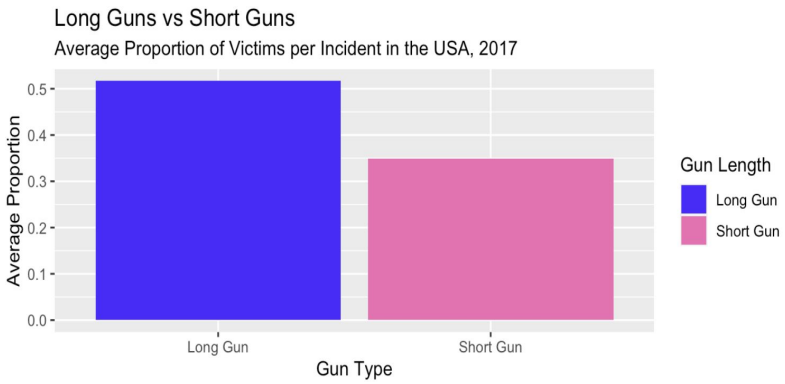
Conclusion: Our analysis showed that long barrel guns cause statistically higher proportion of casualties per incident.

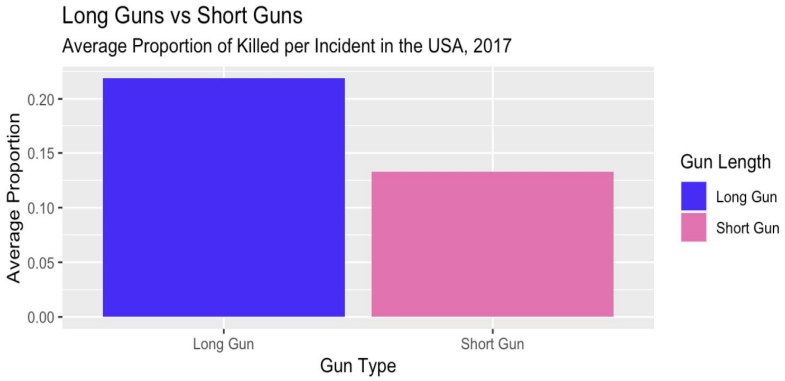
Question 2: Is there a statistically significant difference in the proportion of casualties between stolen guns and guns that were not reported as stolen?

Conclusion: We found that guns that were not reported as stolen cause statistically higher casualty rates than stolen guns.

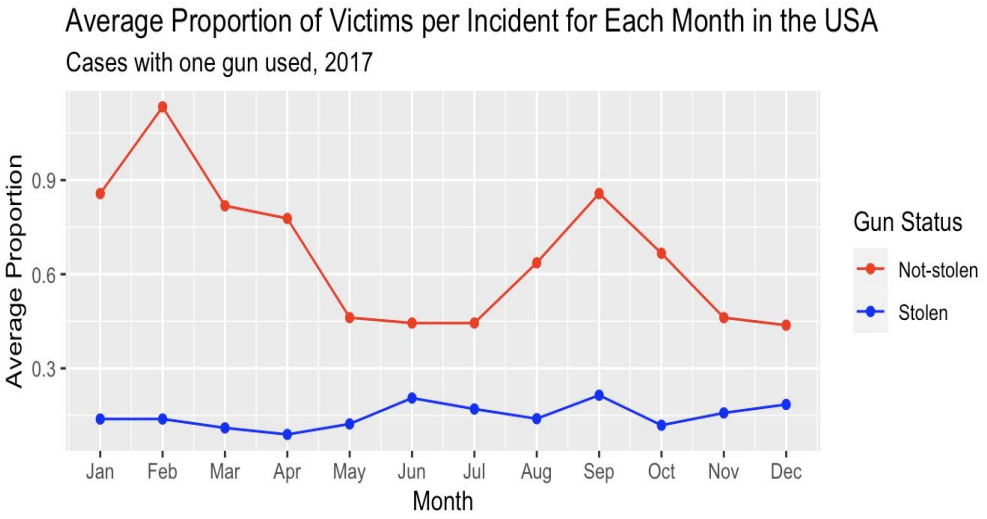
Graphing of Question 1

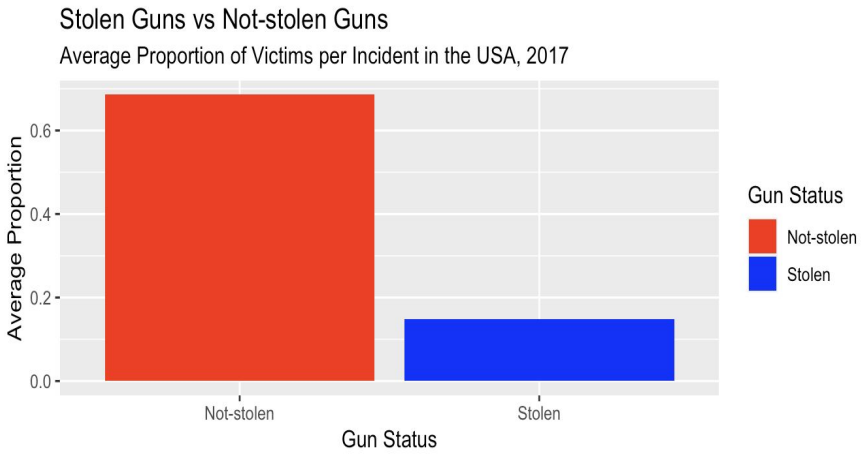






Graphing of Question 2





Conclusions & Interpretation

- **Gun Types (Long vs Short) -** This analysis reveals that incidents involving long guns result in higher casualty rates compared to incidents involving short guns. Long guns, which include rifles, shotguns etc exhibit a greater chance for causing fatalities and injuries in gun violence incidents.

- **Gun Status (Stolen vs not Stolen) -** This analysis demonstrates that incidents involving not stolen guns lead to higher casualty rates than incidents involving stolen guns. Firearms reported as legal possession are associated with increased lethality in gun violence incidents, indicating a correlation between firearm ownership status and the severity of outcomes.